

# Spectral Theory and Hilbert-Space Embedding of the FFGFT Tori

Mathematical Foundation of the  $T^4/\mathbb{Z}_3$  Spectral Mechanics

## Contents

|    |                                                                  |   |
|----|------------------------------------------------------------------|---|
| .1 | Foundations of Spectral Theory in Hilbert Spaces . . . . .       | 2 |
|    | Motivation: from matrices to operators . . . . .                 | 2 |
|    | The Hilbert space: definition and physical meaning . . . . .     | 3 |
|    | The spectrum of a linear operator . . . . .                      | 3 |
|    | The spectral theorem: centrepiece of the theory . . . . .        | 3 |
| .2 | The Fundamental Fractal Operator $\hat{F}$ . . . . .             | 3 |
|    | Definition and Hilbert-space structure . . . . .                 | 3 |
|    | $\xi$ as smallest eigenvalue of the $D_4$ sub-operator . . . . . | 4 |
| .3 | The FFGFT State Space . . . . .                                  | 4 |
|    | Tensor-product structure . . . . .                               | 4 |
|    | The fractal measure $d\mu_f$ with 100-fold recursion . . . . .   | 4 |
| .4 | The Torus $T^4$ in Hilbert Space . . . . .                       | 4 |
|    | Fourier decomposition on $T^4$ . . . . .                         | 4 |
|    | The $\mathbb{Z}_3$ orbifold projection . . . . .                 | 5 |
|    | Fixed points and neutrino localisation . . . . .                 | 5 |
| .5 | The Complete Spectral Operator . . . . .                         | 5 |
|    | Tensor-product form . . . . .                                    | 5 |
|    | Commutator conditions and MASA . . . . .                         | 5 |
| .6 | Generalised Fourier Transform on $T^4/\mathbb{Z}_3$ . . . . .    | 5 |
| .7 | Spectral Derivation of Particle Masses . . . . .                 | 5 |
|    | Charged leptons as torus excitations . . . . .                   | 5 |
|    | Neutrinos as fixed-point eigenstates . . . . .                   | 6 |
|    | Spectral selection rule for the exponent ladder . . . . .        | 6 |
| .8 | Fractal Dimension and $K_{\text{frac}}$ . . . . .                | 6 |
| .9 | Complete Spectral Correspondence Table . . . . .                 | 6 |

## List of Symbols

| Symbol                 | Meaning                                  | Value / Unit                     |
|------------------------|------------------------------------------|----------------------------------|
| $\mathcal{H}$          | Hilbert space of physical states         | separable                        |
| $\hat{F}$              | fundamental fractal spectral operator    | self-adjoint                     |
| $\hat{F}_{D_4}$        | $D_4$ sub-operator                       | eigenvalue $\xi$                 |
| $\sigma(\hat{F})$      | spectrum of $\hat{F}$                    | $\subset \mathbb{R}$             |
| $E(\lambda)$           | spectral projection measure              | $E(\lambda)^2 = E(\lambda)$      |
| $\mathcal{D}(\hat{F})$ | domain of $\hat{F}$                      | dense in $\mathcal{H}$           |
| $P_{\mathbb{Z}_3}$     | $\mathbb{Z}_3$ projection operator       | idempotent                       |
| $\hat{\Delta}_{T^4}$   | Laplacian on $T^4/\mathbb{Z}_3$          | essentially self-adj.            |
| $\hat{W}$              | winding-number operator                  | eigenvalues $(n_\theta, n_\phi)$ |
| $D_f$                  | fractal dimension of the orbifold        | $3 - \xi$                        |
| $D_f^{\text{frac}}$    | approximate fractal dimension            | 2.94 (no $\xi$ -derivation)      |
| $K_{\text{frac}}$      | interface factor rolled $\rightarrow$ SI | $1 - 100\xi = 74/75$ (Doc. 133)  |
| $\xi$                  | FFGFT geometry parameter                 | $4/30000$                        |
| $\omega$               | $\mathbb{Z}_3$ phase factor              | $e^{2\pi i/3}$                   |
| [K]                    | derivable from $\xi$                     | status marker                    |
| [B]                    | algebraically proved                     | status marker                    |
| [S]                    | sketched / plausible                     | status marker                    |
| [SETZUNG]              | axiomatic setting                        | status marker                    |

## Summary

Doc. 322 develops the mathematical foundation of FFGFT spectral theory. The mass formulae of Doc. 320 receive here their rigorous Hilbert-space embedding: the fundamental fractal operator  $\hat{F}$  on the state space  $\mathcal{H}_{\text{FFGFT}}$  has the particle masses as eigenvalues. The geometry parameter  $\xi = 4/30000$  is the smallest eigenvalue of the  $D_4$  sub-operator. The mass formulae of Docs. 006, 317 and 320 are not phenomenological ansätze but eigenvalue equations of a self-adjoint operator.

### Status markers

All statements carry FFGFT status markers:

**[K]** derivable from  $\xi$    **[B]** algebraically proved   **[S]** sketched/plausible   **[SETZUNG]** axiomatic setting

## .1 Foundations of Spectral Theory in Hilbert Spaces

### Motivation: from matrices to operators

In linear algebra a symmetric  $n \times n$  matrix  $A$  possesses exactly  $n$  real eigenvalues and an orthonormal eigenbasis, enabling the diagonal decomposition  $A = UDU^T$ .

In quantum mechanics – and in FFGFT – operators act on *infinite-dimensional* Hilbert spaces. The momentum operator, the Hamiltonian, and the FFGFT spectral operator have no finite-dimensional analogue. Spectral theory generalises the eigenvalue concept to this case.

## The Hilbert space: definition and physical meaning

**Definition (Hilbert space).** A Hilbert space  $\mathcal{H}$  is a complete inner-product vector space. Its elements have finite norm  $\|\psi\| = \sqrt{\langle\psi, \psi\rangle} < \infty$ , which in position space means

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx < \infty. \quad (1)$$

Only such square-integrable wave functions are physically realisable – they describe normalisable probability densities.

**Note:** Eigenstates of operators with *continuous* spectrum – e.g. position eigenstates  $|y\rangle$  with  $\langle x|y\rangle = \delta(x - y)$  – are not normalisable and lie outside  $\mathcal{H}$  itself, in the extended distributional space. In FFGFT this issue arises in the IR limit; the sub-Planck regime has a purely discrete spectrum.

## The spectrum of a linear operator

For a linear operator  $T$  on a Hilbert space the spectrum  $\sigma(T)$  generalises the set of all eigenvalues:

$$\sigma(T) = \{\lambda \in \mathbb{C} : (T - \lambda I) \text{ not invertible with bounded inverse}\}. \quad (2)$$

One distinguishes:

- **Point spectrum**  $\sigma_p(T)$ : “true” eigenvalues with  $T\psi = \lambda\psi$ ,  $\psi \neq 0$ .
- **Continuous spectrum**  $\sigma_c(T)$ :  $(T - \lambda I)$  injective but not surjective; no normalised eigenvector.
- **Absolutely continuous and singular spectrum**: important for Schrödinger operators on fractal spaces.

## The spectral theorem: centrepiece of the theory

**[B]** For self-adjoint operators ( $T = T^\dagger$  with suitable domain) the spectral theorem holds: there exists a unique *spectral measure*  $E$  – a projection-valued function  $\lambda \mapsto E(\lambda)$  – such that

$$T = \int_{\sigma(T)} \lambda dE(\lambda). \quad (3)$$

For compact operators Eq. (3) reduces to the familiar discrete sum  $T = \sum_i \lambda_i \langle \cdot, e_i \rangle e_i$ . Self-adjointness guarantees  $\sigma(T) \subset \mathbb{R}$  – real spectra, the necessary condition for measurable observable values.

## .2 The Fundamental Fractal Operator $\hat{F}$

### Definition and Hilbert-space structure

**[SETZUNG]** FFGFT postulates a quantum-logical fractal operator  $\hat{F}$  acting on a separable Hilbert space  $\mathcal{H}$ , formally encoding the fractal self-similarity of the sub-Planck geometry:

$$\hat{F} \equiv \bigoplus_{n=1}^{\infty} S_n \circ L_n, \quad (4)$$

with scale operators  $S_n = r_n U_n$  ( $r_n \in (0, 1)$  contraction rates,  $U_n$  unitary) and logical projectors  $L_n$  satisfying  $L_n^2 = L_n$ ,  $L_n^\dagger = L_n$ ,  $L_n L_m = L_{\min(n,m)}$ .

The domain

$$\mathcal{D}(\hat{F}) = \left\{ \psi \in \mathcal{H} : \sum_{n=1}^{\infty} \|S_n(L_n \psi)\|^2 < \infty \right\} \quad (5)$$

is dense in  $\mathcal{H}$ . **[B]**  $\hat{F}$  is self-adjoint: since  $L_0 = \xi \cdot \ell_P$  (Doc. 180) renders the scale ladder finite,  $\sum_n r_n < \infty$  and hence  $\hat{F}$  is bounded with  $\mathcal{D}(\hat{F}) = \mathcal{H}$ ; symmetry follows from the  $\mathbb{Z}_3$  pairing  $(k, -k)$ . The deficiency indices are  $(0, 0)$  – there is exactly one self-adjoint realisation (Doc. 327, R86).

### $\xi$ as smallest eigenvalue of the $D_4$ sub-operator

**[K]** The geometry parameter is not a free parameter but a spectral value:

$$\xi = \lambda_{\min}(\hat{F}_{D_4}) = \frac{4}{30000}. \quad (6)$$

This explains structurally why  $\xi$  appears as the base of all mass formulae: it is not a chosen parameter but the lowest mode of the geometry operator.

## .3 The FFGFT State Space

### Tensor-product structure

**[K]** The physical state space:

$$\mathcal{H}_{\text{FFGFT}} = \mathcal{H}_{\text{geom}} \otimes \mathcal{H}_{\text{spin}} \otimes \mathcal{H}_{\text{flavor}}, \quad (7)$$

with:

1.  $\mathcal{H}_{\text{geom}} = L^2(T^4/\mathbb{Z}_3, d\mu_f)$ : square-integrable functions on the fractal orbifold.
2.  $\mathcal{H}_{\text{spin}} = \mathbb{C}^{2[D_f]}$ : spinor space;  $[D_f] = 3$ .
3.  $\mathcal{H}_{\text{flavor}} = \ell^2(\mathbb{N}^2)$ : flavour space with winding numbers  $(n_\theta, n_\phi)$ .

### The fractal measure $d\mu_f$ with 100-fold recursion

**[K]** The fractal measure on  $T^4/\mathbb{Z}_3$  is built by 100 successive fractalisations of the Lebesgue measure:

$$d\mu_f^{(0)}(x) = dx, \quad (8)$$

$$d\mu_f^{(k+1)}(x) = \prod_{i=1}^4 \left( \sum_{n=1}^{100} c_n^{(k)} \delta(x_i - x_i^{(n)}) \right) d\mu_f^{(k)}(x), \quad (9)$$

with  $c_n^{(k)} = n^{-(D_f^{(k)}-3)}$ ,  $D_f^{(k)} = 3 - \xi^{(k)}$ ,  $\xi^{(k)} = (4/30000)(1 - k/100)$ . After 100 steps:

$$\boxed{d\mu_f(x) = \prod_{i=1}^4 \left( \sum_{n=1}^{100} n^\xi \delta(x_i - x_i^{(n)}) \right) dx_i}, \quad \xi = \frac{4}{30000}. \quad (10)$$

## .4 The Torus $T^4$ in Hilbert Space

### Fourier decomposition on $T^4$

**[B]**

$$L^2(T^4) \cong \bigoplus_{n \in \mathbb{Z}^4} \mathbb{C} \cdot e^{in \cdot x/R}, \quad \Delta_{T^4} e^{in \cdot x/R} = -\frac{|n|^2}{R^2} e^{in \cdot x/R}. \quad (11)$$

## The $\mathbb{Z}_3$ orbifold projection

[K]

$$P_{\mathbb{Z}_3} = \frac{1}{3} \sum_{k=0}^2 \omega^k \tau^k, \quad \mathcal{H}(T^4/\mathbb{Z}_3) = \text{Fix}(P_{\mathbb{Z}_3}) \subset L^2(T^4). \quad (12)$$

## Fixed points and neutrino localisation

[K] The boundary condition at a fixed point  $x_0$ :

$$\psi_\chi(x_0 + y) = \chi \cdot \psi_\chi(x_0 + g_*(y)). \quad (13)$$

The three neutrino flavours are localised at fixed points (Doc. 320):

$$\nu_1 \leftrightarrow F_2 = \frac{2\pi R}{3}(1, 1, 1, 0), \quad \nu_2 \leftrightarrow F_5 = \frac{4\pi R}{3}(1, 1, 1, 0), \quad \nu_3 \leftrightarrow F_7 = \frac{4\pi R}{3}(1, 1, 1, 1). \quad (14)$$

## .5 The Complete Spectral Operator

### Tensor-product form

[K]

$$\hat{F} = \hat{\Delta}_{T^4} \otimes \hat{S} \otimes \hat{W}, \quad \hat{M}_f = v \cdot \xi \hat{P} \otimes \hat{r}. \quad (15)$$

### Commutator conditions and MASA

[K]

$$[\hat{F}, \hat{S}] = 0, \quad [\hat{n}_\theta, \hat{n}_\phi] = 0. \quad (16)$$

The MASA  $\mathcal{A} = \text{span}\{\hat{n}_\theta, \hat{n}_\phi, \hat{p}\}$  yields the complete spectral labelling  $(n_\theta, n_\phi, p)$ .

## .6 Generalised Fourier Transform on $T^4/\mathbb{Z}_3$

[K] Projection kernel and GFT:

$$K_\lambda^{(p)}(x) = \sum_{n \in \mathbb{Z}^4} \frac{\delta_{\lambda, \Lambda_n^{(p)}}}{|n|^{D_f/2}} e^{in \cdot x/R}, \quad \Lambda_n^{(p)} = \frac{|n|^2}{R^2} \cdot \xi p. \quad (17)$$

$$\tilde{\psi}(\lambda) = \langle K_\lambda^{(p)}, \psi \rangle_{\mathcal{H}} = \int_{T^4/\mathbb{Z}_3} \overline{K_\lambda^{(p)}(x)} \psi(x) d\mu_f(x). \quad (18)$$

## .7 Spectral Derivation of Particle Masses

### Charged leptons as torus excitations

[K] The mass eigenvalue equation  $\hat{M}_{e,\mu,\tau}|n_\theta, n_\phi, p\rangle = m_i|n_\theta, n_\phi, p\rangle$  gives:

$$m_i = \frac{n_\phi^2}{n_\theta} \cdot \xi^{p_i} \cdot v. \quad (19)$$

| Particle     | $n_\theta$ | $n_\phi$ | $p_i$ | $m_i$      |
|--------------|------------|----------|-------|------------|
| Electron $e$ | 3          | 2        | 3/2   | 0.511 MeV  |
| Muon $\mu$   | 5          | 4        | 1     | 104.96 MeV |
| Tau $\tau$   | 9          | 5        | 2/3   | 1783.5 MeV |

## Neutrinos as fixed-point eigenstates

[K]

$$P_{\mathbb{Z}_3}|v_i\rangle = |v_i\rangle, \quad m_{v_i} = m_e \cdot \xi^{p_i}. \quad (20)$$

| Neutrino | Fixed point | $\dim_{\text{loc}}$ | $p_i$ | $m_{v_i}$ |
|----------|-------------|---------------------|-------|-----------|
| $\nu_1$  | $F_2$       | $2 - 1/4$           | $9/4$ | 0.976 meV |
| $\nu_2$  | $F_5$       | 2                   | 2     | 9.084 meV |
| $\nu_3$  | $F_7$       | $2 - 1/5$           | $9/5$ | 54.11 meV |

## Spectral selection rule for the exponent ladder

[K]

$$\Delta p = \frac{k}{3}, \quad k \in \mathbb{Z}, \quad \hat{F} \cdot \hat{\xi}^{1/3} = \hat{\xi}^{1/3} \cdot \hat{F}. \quad (21)$$

## .8 Fractal Dimension and $K_{\text{frac}}$

[K] Exact FFGFT formula:

$$D_f = 3 - \xi = 3 - \frac{4}{30000} = \frac{22499}{7500} \approx 2.9998\bar{6}. \quad (22)$$

Doc. 133 gives the interface factor:

$$K_{\text{frac}} = 1 - 100\xi = \frac{74}{75} \approx 0.9866\bar{6}. \quad (23)$$

In mass ratios  $K_{\text{frac}}$  cancels exactly (Doc. 012).

## .9 Complete Spectral Correspondence Table

**Table 1:** All spectral states of the FFGFT and their Hilbert-space components (Doc. 322)

| Sector    | State           | Eigenvalue | Formula                    | $\mathcal{H}$ -component        | Status |
|-----------|-----------------|------------|----------------------------|---------------------------------|--------|
| Leptons   | $ e\rangle$     | 0.511 MeV  | $\frac{4}{3}\xi^{3/2}\nu$  | $\mathcal{H}_{\text{fl}}(3, 2)$ | [K]    |
|           | $ \mu\rangle$   | 104.96 MeV | $\frac{16}{5}\xi^1\nu$     | $\mathcal{H}_{\text{fl}}(5, 4)$ | [K]    |
|           | $ \tau\rangle$  | 1783.5 MeV | $\frac{25}{9}\xi^{2/3}\nu$ | $\mathcal{H}_{\text{fl}}(9, 5)$ | [K]    |
| Neutrinos | $ \nu_1\rangle$ | 0.976 meV  | $m_e \xi^{9/4}$            | $\text{Fix}(P) \cap F_2$        | [K]    |
|           | $ \nu_2\rangle$ | 9.084 meV  | $m_e \xi^2$                | $\text{Fix}(P) \cap F_5$        | [K]    |
|           | $ \nu_3\rangle$ | 54.11 meV  | $m_e \xi^{9/5}$            | $\text{Fix}(P) \cap F_7$        | [K]    |
| Bosons    | $ g_a\rangle$   | 0          | $\hat{F} g_a\rangle = 0$   | $\text{Fix}(P_{\mathbb{Z}_3})$  | [K]    |
|           | $ W^\pm\rangle$ | 80.38 GeV  | $\frac{1}{2}g_2\nu$        | $\mathcal{H}_{\text{spin}}$     | [B]    |
|           | $ Z^0\rangle$   | 91.19 GeV  | $m_W/\cos\theta_W$         | $\mathcal{H}_{\text{spin}}$     | [B]    |

**Overall status Doc. 322**

The Hilbert-space embedding of FFGFT is structurally complete:  $\hat{F}$  is defined, the state space is constructed, all mass formulae from Doc. 320 are identified as eigenvalue equations, and the self-adjointness of  $\hat{F}$  is proved (Doc. 327, R86):  $\xi = \lambda_{\min}(\hat{F}_{D_4})$  is thereby well-defined as a spectral value. The particle spectrum follows from spectral theory, not from free parameters.